

1 Differential Evolution (DE) Optimization Converge

In general, Differential Evolution (DE) optimization can converge to different local minima depending on the non-linear nature of the objective function. A common approach in nonlinear optimization problems, is to perform multiple optimization runs using different initial values of the decision variables (also referred to as optimization parameters). We do the same for the DE procedure, with different initial values for our decision variables R_a , R_b , and T_1 . To illustrate, we randomly select 10 different sets of initial decision variable values from the defined search spaces. These sets are used as starting points for the DE optimization runs for both Th_1 and Th_2 ¹. The search spaces considered in this work are $[0.5, 1000]$ k Ω for both R_a and R_b , and $[1, 200]$ ms for T_1 .

Table 1 and Table 2 summarizes the selected initial decision variable values, the optimized decision variable values obtained from DE, and the resulting objective function values for each optimization run.²

DE Run Count	Initial Values of Decision Variables: (R_{ai} , R_{bi} , T_{1i})			DE Optimized Decision Variables & Objective Function Values			
	R_{ai} (k Ω)	R_{bi} (k Ω)	T_{1i} (ms)	R_a (k Ω)	R_b (k Ω)	T_1 (ms)	%NL
Run1	0.5	0.5	1	16.44	0.615	5.647	0.2203
Run2	10	10	2	16.44	0.615	5.646	0.2205
Run3	100	100	4	16.43	0.615	5.645	0.2205
Run4	200	200	8	16.44	0.615	5.646	0.2203
Run5	300	300	16	16.51	0.617	5.642	0.2206
Run6	400	400	32	16.4	0.613	5.648	0.2206
Run7	500	500	64	16.43	0.615	5.646	0.2205
Run8	700	700	128	16.4	0.613	5.649	0.2206
Run9	900	900	180	16.45	0.615	5.648	0.2204
Run10	1000	1000	200	16.44	0.615	5.646	0.2205

Table 1: Illustration of the optimized % NL values obtained from DE procedure with random initialization. The thermistor Th_1 is used for the temperature range of $[0, 120]$ °C.

DE Run Count	Initial Values of Decision Variables R_{ai} , R_{bi} , T_{1i}			DE Optimized Decision Variables & Objective Function Values			
	R_{ai} (k Ω)	R_{bi} (k Ω)	T_{1i} (ms)	R_a (k Ω)	R_b (k Ω)	T_1 (ms)	%NL
Run1	0.5	0.5	1	14.31	0.5	5.545	0.3245
Run2	10	10	2	14.28	0.501	5.538	0.3257
Run3	100	100	4	14.26	0.5	5.544	0.3257
Run4	200	200	8	14.3	0.5	5.543	0.3259
Run5	300	300	16	14.28	0.5	5.543	0.3258
Run6	400	400	32	14.31	0.5	5.545	0.3245
Run7	500	500	64	14.25	0.5	5.543	0.3255
Run8	700	700	128	14.26	0.5	5.544	0.3257
Run9	900	900	180	14.25	0.5	5.543	0.3255
Run10	1000	1000	200	14.31	0.5	5.546	0.3249

Table 2: Illustration of the optimized % NL values obtained from DE procedure with random initialization. The thermistor Th_2 is used for the temperature range of $[0, 120]$ °C.

¹ Th_1 and Th_2 refer to thermistors with nominal resistance values of 12K Ω and 10K Ω , respectively

²In the table 1 and table 2, R_{ai} , R_{bi} , and T_{1i} denote the initial decision variable values used for initialization of the DE optimization process, where the subscript i denotes the initial values selected for a particular optimization run. These values correspond to the starting points assigned to the decision variables R_a , R_b , and T_1 , respectively, before the optimization begins. For each set of initial decision variable values, the DE algorithm iteratively searches within the defined search space and generates a corresponding set of optimized decision variable values, namely R_a , R_b , and T_1 , upon convergence. These optimized values are the DE optimized decision variables. The objective function value corresponding to these optimized decision variable values represents the final performance measure obtained after convergence of the DE optimization algorithm.

It can be observed from Tables 1 and 2, that although the optimized decision variable values vary slightly for different initializations, the obtained percentage nonlinearity (%NL) changes only within a narrow range of 0.2203% to 0.2206% and 0.3245% to 0.3259% for thermistors, Th_1 and Th_2 , respectively. Hence, although DE may converge to different local minima, the extremely small variation in the obtained objective-function values across multiple independent runs provides strong empirical evidence for the robustness of the DE procedure for the optimization problem considered in the paper. While global optimality cannot be theoretically guaranteed, the repeated convergence to nearly identical objective-function values demonstrates the robustness and reliability of the proposed optimization approach.

2 Computational Complexity and Runtime of DE Optimization

The DE procedure maintains a population of N_p decision variable vectors. For each such vector, DE evaluates the *fitness* by finding out the objective function value % NL and also evaluates the following constraints: (i) the conversion time, $T_C \leq 200$ ms, (ii) the conditions, $R_a > R_b$, and $\delta \in [0.1, 2]$ must be satisfied. The DE procedure then iterates for a maximum of G_{max} steps. If C denotes the complexity of evaluating the objective function and the constraints for a single decision variable vector, then the total computational complexity of DE is $\mathcal{O}(N_p G_{max} N)$.

Now, we theoretically characterize the computational complexity C (which is 906 mathematical operations in this work for $N = 25$) of evaluating the objective function and the constraints for one element of the population set in the DE procedure. We note that evaluating the objective function (which is % NL) requires

1. finding out the response to each measurement point,
2. finding the best-fit line using all measurements,
3. then, the computation of % NL.

For the following discussion, let N denote the number of data points. For each data point, we note that the response is $D_{\theta_k} = \frac{T_{\theta_k}}{T_2}$, θ_k is a temperature and the corresponding resistance is R_{θ_k} , k is an index with $k \in \{1, 2, \dots, N\}$, where

$$T_{\theta_k} = T_1 \left(\frac{R_{\theta_k}}{R_{\theta_k} + R_a} \right) + T_2 \left(\frac{R_{\theta_k}}{R_{\theta_k} + R_b} \right). \quad (1)$$

Thus, computation of these values requires $\mathcal{O}(N)$ operations. Then, we need to obtain the best fit line: $y = m\theta_k + c$, which again requires $\mathcal{O}(N)$ operations. Finally, computation of the % NL also requires $\mathcal{O}(N)$ operations. The computation of the constraints are also linear in N . So in every step, the computational complexity is $\mathcal{O}(N)$. Therefore, the complexity of the DE procedure is $\mathcal{O}(N_p G_{max} N)$.

We note that practically, the number of iterations required could be much less than To illustrate, we evaluate the practical computational performance of the DE algorithm for multiple independent optimization runs with both Th_1 and Th_2 . The DE algorithm implementation in the Python SciPy library was used. The computing system is equipped with a 12th Gen Intel(R) Core(TM) i5-1235U processor with a 64-bit architecture, 10 cores (2 performance cores and 8 efficient cores), and 12 threads, a maximum turbo frequency of 4.40 GHz, and 12 MB of Intel Smart Cache. The DE algorithm is terminated under the following conditions: (i) when the maximum number of iterations (= 1000) is reached, or (ii) when the convergence tolerance is satisfied, i.e., when the standard deviation of the population objective values falls below the threshold $atol + tol \cdot |\text{mean}(f)|$, where tol (= 0.01) and $atol$ (= 0) are the relative and absolute tolerances, respectively. We obtain number of iterations (G), the total number of objective function evaluations (NFEV), runtime (in seconds)³, optimal decision variables (R_a , R_b , T_1), and the corresponding objective function value (%NL). These results are summarized in Tables 3 and 4, respectively. For example, in Table 3 corresponding to Th_1 , Run9 converged after 93 iterations with a total of 4061 objective function evaluations and a practical computational effort of approximately 3.68 million mathematical operations in 0.517 s. The runtime varied⁴ across

different runs depending on the number of iterations required for convergence, the number of objective function evaluations, and the corresponding computational effort, as reported in Tables 3 and 4. Therefore, although the theoretical maximum number of mathematical operations is 13.59 million (based on full utilization of the maximum iterations and population size), the practical computational effort for Run9 (Th_1) is approximately 3.68 million operations (based on 4061 function evaluations and an estimated 906 operations per evaluation). This reduction reflects early termination of the optimization process under the convergence criteria and the stochastic nature of the DE algorithm. The practical computational time (< 0.6 seconds in offline mode) required for the DE procedure is therefore *reasonable*.

DE Run Count	Number of Iterations (G_{Runi})	Number of objective function evaluations ($NFEV_{Runi}$)	Practical Number of Mathematical Operations in Millions ($MAOP_{Runi}$)	Theoretical Maximum Number of Mathematical Operations in Millions ($MAOP_{max}$)	DE Runtime (s)	R_a k Ω	R_b k Ω	T_1 (ms)	%NL
Run1	76	3263	2.956278	13.59	0.586	16.4	0.614	5.647	0.2206
Run2	67	2805	2.54133	13.59	0.41	16.43	0.615	5.646	0.2205
Run3	83	3569	3.233514	13.59	0.483	16.47	0.615	5.643	0.2204
Run4	95	4175	3.78255	13.59	0.531	16.43	0.615	5.646	0.2205
Run5	80	3430	3.10758	13.59	0.471	16.41	0.614	5.643	0.2203
Run6	90	3846	3.484476	13.59	0.511	16.42	0.614	5.643	0.2204
Run7	90	3855	3.49263	13.59	0.512	16.43	0.615	5.646	0.2205
Run8	77	3312	3.000672	13.59	0.417	16.4	0.613	5.649	0.2206
Run9	93	4061	3.679266	13.59	0.517	16.42	0.614	5.644	0.2204
Run10	88	3750	3.3975	13.59	0.475	16.4	0.613	5.649	0.2206
1. i refers to i^{th} DE run, where $i \in [1, 10]$ 2. s and ms refer to seconds and milliseconds, respectively. 3. $MAOP_{Runi}$ ($= C \cdot NFEV_{Runi}$) represents the practical number of mathematical operations for the i^{th} run of the DE algorithm, based on the number of objective function evaluations $NFEV_{Runi}$ 4. $MAOP_{max}$ ($= C \cdot N_p \cdot G_{max} = 906 \cdot 15 \cdot 1000 = 13.59$ million) represents the theoretical maximum number of mathematical operations for the DE algorithm, based on the population size ($=15$) and maximum number of iterations ($=1000$). $MAOP_{max}$ is the same for all runs since C , N_p , and G_{max} are fixed in this work.									

Table 3: Computational Complexity Analysis of DE Algorithm with Thermistor, Th_1 for the temperature range of $[0, 120]$ °C.

³DE runtime is defined as the elapsed wall-clock time between the initiation and termination of the DE procedure, measured using `time.perf_counter()` in Python. It includes the time spent on objective function evaluations as well as internal evolutionary operations such as mutation, crossover, selection, and constraint handling.

⁴The runtime variations observed across the 10 runs reported in Table 3 (for Th_1) and Table 4 (for Th_2) should be interpreted with caution; due to the stochastic nature of the optimization algorithm, runtime may vary depending on initialization and the optimization trajectory followed in each run. Consequently, differences in runtime both within and between the Th_1 and Th_2 experiments cannot be attributed solely to the number of iterations and the number of objective function evaluations.

DE Run Count	Number of Iterations (G_{Runi})	Number of objective function evaluations ($NFEV_{Runi}$)	Practical Number of Mathematical Operations in Millions ($MAOP_{Runi}$)	Theoretical Maximum Number of Mathematical Operations in Millions ($MAOP_{max}$)	DE Runtime (sec)	R_a k Ω	R_b k Ω	T_1 (ms)	%NL
Run1	80	3019	2.735214	13.59	0.438	14.26	0.5	5.545	0.3269
Run2	100	3962	3.589572	13.59	0.534	14.25	0.5	5.541	0.3266
Run3	75	2872	2.602032	13.59	0.398	14.28	0.501	5.54	0.3262
Run4	82	2982	2.701692	13.59	0.424	14.42	0.5	5.555	0.3269
Run5	80	2880	2.60928	13.59	0.421	14.3	0.5	5.544	0.3262
Run6	74	2704	2.449824	13.59	0.393	14.3	0.5	5.544	0.3262
Run7	92	3243	2.938158	13.59	0.464	14.28	0.5	5.544	0.3267
Run8	81	3093	2.802258	13.59	0.432	14.27	0.501	5.537	0.3262
Run9	90	3478	3.151068	13.59	0.492	14.25	0.501	5.538	0.3262
Run10	98	3898	3.531588	13.59	0.525	14.25	0.501	5.537	0.3264
1. i refers to i^{th} DE run, where $i \in [1, 10]$ 2. s and ms refer to seconds and milliseconds, respectively. 3. $MAOP_{Runi}$ ($= C \cdot NFEV_{Runi}$) represents the practical number of mathematical operations for the i^{th} run of the DE algorithm, based on the number of objective function evaluations $NFEV_{Runi}$ 4. $MAOP_{max}$ ($= C \cdot N_p \cdot G_{max} = 906 \cdot 15 \cdot 1000 = 13.59$ million) represents the theoretical maximum number of mathematical operations for the DE algorithm, based on the population size ($=15$) and maximum number of iterations ($=1000$). $MAOP_{max}$ is the same for all runs since C, N_p, and G_{max} are fixed in this work.									

Table 4: Computational Complexity Analysis of DE Algorithm with Thermistor, Th_2 for the temperature range of $[0, 120]^\circ\text{C}$.